



RESEARCH INTERESTS

Computer Vision, Machine Learning Pattern Recognition, Deep Learning, Reinforcement Learning, Generative Models, Reinforcement Learning, Software Engineering

EDUCATION

- **Bachelor of Computer Engineering** 2022 - 2026
Islamic Azad University | **GPA: 4/4 - 20 (out of 20)**
Ranked 2nd in Iran according to **usnews** ranking

Highlighted Courses

- | | |
|---|--|
| ○ Analysis Of Algorithms – Grade: 4/4
by Dr. MohammadReza Sarshar | ○ Operating Systems – Grade: 4/4
by Dr. |
| ○ Operating Systems – Grade: 4/4
by Dr. | ○ Data Base – Grade: 4/4
by Dr. Niloofar Mirzaee |
| ○ نظريه زبان ها و ماشين ها – Grade: 4/4
by Dr. MohammadReza Sarshar | ○ شبكه – Grade:
by Dr. Hessam Edrisi |

- **High School – Diploma of Mathematics** 2019 - 2022
Shahid Soltani Talented Students School (Sampad) | **GPA: 4/4 - 20 (out of 20)**
Ranked 1st in Iran - Shahid Soltani is among nationwide schools developed to provide education for Iran's **exceptionally talented** students and are managed by the National Organization for Development of Exceptional Talents.

RESEARCH EXPERIENCE

Human Pose Estimation for Automated Exercise Movement Assessment Using Computer Vision - 2026

Designed and implemented a computer vision system for automated analysis of exercise movements using human pose estimation techniques.

- Using MediaPipe to extract body landmarks and track joint coordinates across video frames for different types of exercises with different types of configurations for each one.
- Developed rule-based algorithms for repetition counting based on joint angle thresholds and motion trajectories.
- **Addressed challenges:** such as pose landmark noise, variation in camera angle, and movement inconsistency through smoothing techniques.
- Achieved 90% - 95% accuracy in counting user's repetitions.
- Implemented a web-based backend using Django and Django REST Framework for storing and managing user data.

Supervisor: Dr. Sarshar

PROJECTS

- **Real-Time Face Recognition Using Feature Embedding Techniques**
 - Implemented a real-time face recognition system using **OpenCV** and the **face_recognition** library.
 - Constructed a labeled dataset (~30 images per individual) and generated facial feature embeddings for identity matching.
 - Designed an encoding comparison pipeline for recognizing known individuals and labeling unknown faces; Also, Integrated webcam-based live detection and identification.
- **Employee Attendance Management System**
 - Developed a web-based employee attendance and request management platform.
 - Designed and implemented backend architecture using Python, Django, DRF, and PostgreSQL, developing ~30 RESTful API endpoints for an internal employee attendance platform.
 - Designed an administrative dashboard allowing managers to monitor employee working hours and monthly reports.
- **Retrieval-Augmented Generation (RAG) System for Persian (farsi)**
 - Developing a RAG-based system that integrates large language models with external knowledge sources with almost 95% accuracy in retrieving related document and pass it to the LLM to answer user's query.
 - Implementing document ingestion, embedding generation, and vector retrieval pipelines using LangChain.
 - Exploring vector databases and retrieval strategies to improve answer accuracy and contextual grounding.

TEACHING ASSISTANT

- | | |
|--|---------------------|
| • Basic Programming: <i>by Dr. MohammadReza Sarshar</i> | <i>Season- Year</i> |
| • Basic Programming: <i>by Dr. MohammadReza Sarshar</i> | <i>Season- Year</i> |

WORK EXPERIENCE

Co-Founder & AI / Backend Developer *March 2026 - present*

DEMORA AI & Software Startup, Tehran

- Co-founded a 9-member startup and led backend system design and development, mentoring system architecture.
- Developing a Persian-native Retrieval-Augmented Generation (RAG) system using LangChain and integrating OpenCV-based face recognition for automated attendance verification.

SKILLS SUMMARY

- **Programming Languages:** Python, JavaScript, C++, SQL
- **AI & Machine Learning:** OpenCV, MediaPipe, Numpy, Pandas, Scikit Learn
- **LLM & Retrieval Systems:** LangChain, Retrieval-Augmented Generation (RAG), Vector Databases
- **Backend Development:** Django, Django REST Framework, RESTful API Design
- **Tools:** Git, Docker, Linux Ubuntu, PostgreSQL, MySQL, SQLite, ChromaDB

LANGUAGES

- **English:** TOEFL (Expected: September 2026)

HONORS & AWARDS

- **Top 5 Teams – Coding-Dojo Programming Competition**
Amirkabir University of Technology, Tehran (Hosted by Tapsi)

Ranked among the top 5 teams out of approximately 150 participants in a competitive programming real-world challenge of Tapsi (same as Uber in Iran).

CERTIFICATES

- | | |
|--|---|
| • Deep Learning Specialization
by Andrew Ng | Coursera & DeepLearning.ai – 2026 |
| • CS50’s Web Programming | Harvard University – 2024 |
| • CS50’s Python Programming | Harvard University – 2023 |
| • Prompt Engineering | Quera College, Tehran – 2024 |
| • CS50’s Puzzle Day | Harvard University – 2025 |
| • Generative AI Tools | Sharif University, Tehran (Iran) - 2023 |

REFERENCES

Name Name
Associate professor, Department of Major
Name University, City, Iran
Email: email@univeristy.ac.ir

Name Name
Professor, Faculty of Major
Name University, City, Iran
Email: email@univeristy.ac.ir

Name Name
Associate professor, Department of Major
Name University, City, Iran
Email: email@univeristy.ac.ir

Name Name
Professor, Faculty of Major
Name University, City, Iran
Email: email@univeristy.ac.ir